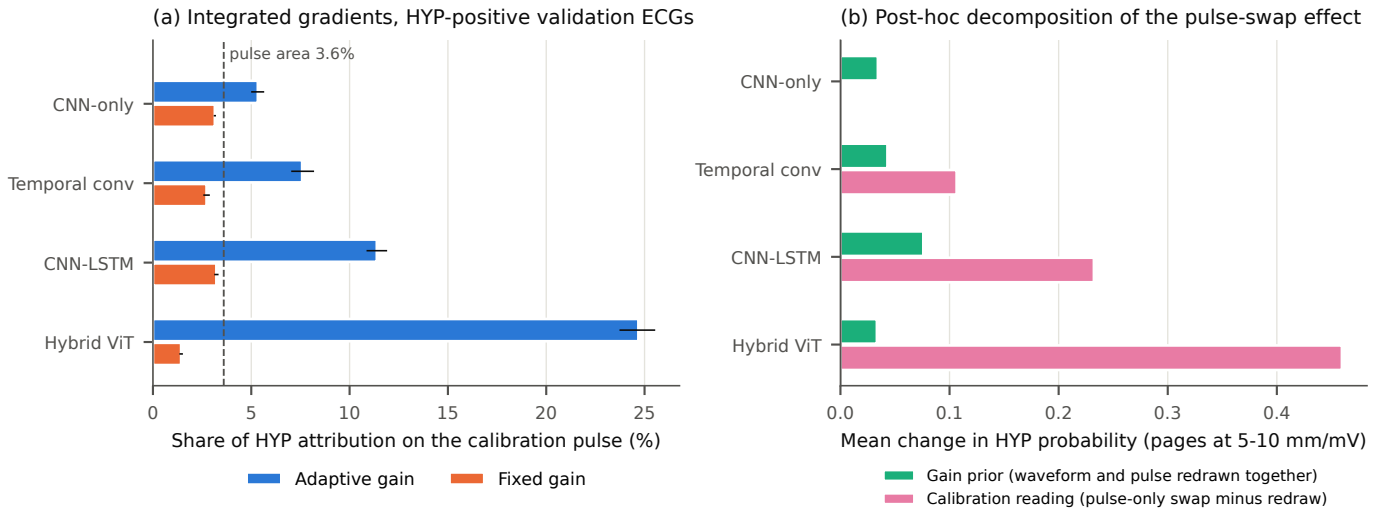
(a) Integrated gradients, HYP-positive validation ECGs
pulse area 3.6%
CNN-only
Temporal conv
CNN-LSTM
Hybrid ViT
0
5
10
15
20
25
Share of HYP attribution on the calibration pulse (%)
Adaptive gain
Fixed gain
(b) Post-hoc decomposition of the pulse-swap effect
CNN-only
Temporal conv
CNN-LSTM
Hybrid ViT
0.0
0.1
0.2
0.3
0.4
Mean change in HYP probability (pages at 5-10 mm/mV)
Gain prior (waveform and pulse redrawn together)
Calibration reading (pulse-only swap minus redraw)